

$$(8 + 11 + 101 + 4)$$

CSE331: Automata and Computability

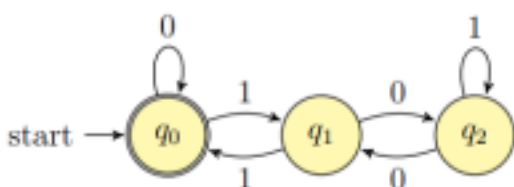
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Question 1

Assignment 1

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((M) = E E gro or more O'S , 12 , 201 * D1 , .]



on $(Q, \Sigma, \delta, q_0, F)$ depicted below

10181E((r)

where $Q = \{q_0, q_1, q_2\}$, $\Sigma = \{0, 1\}$, and $F = \{q_0\}$.

No even 1s 1.81s 1 1.1

a) Find and fix the mistake in the following symbolic description of the transition function δ :

$Q \times \Sigma \rightarrow Q$: for each $j \in \{0, 1\}$

$\delta(q_0, j) = q_j$ $\delta(q_1, j) = q_{1-j}$ $\delta(q_2, j) = q_{1+j}$ b) Keeping the same set of states Q , alphabet Σ , starting state q_0 , and set of accepting states $\{q_0\}$, change the transition function δ so that the resulting finite automaton never

accepts a string ending in a 1.

Modify the Given Machine's &

c) There is a beautiful plain English description of the language recognized by the finite automaton with the state diagram depicted at the start of Question 1. What is it?

Question 2

For each of the languages below, draw the DFA that recognizes the language. You may assume that each language is over the alphabet $\Sigma = \{0, 1\}$.

(a) $\{w \in \Sigma^* \mid \text{the set of all string which do not end with 11}\}$

V

(b) $\{w \in \Sigma^* \mid \text{where the first and last character are the same}\}$

(c) $\{w \in \Sigma^* \mid w \text{ does not have even length and } w \text{ does not contain the string 111}\}$

(d) $\{w \in \Sigma^* \mid \text{if the number of 1's in } w \text{ is odd, then the number of 0's in } w \text{ is also odd}\}$

(e) $\{w \in \Sigma^* \mid w \text{ is any string not in } 0^*1^*\}$

(f) $\{w \in \Sigma^* \mid \text{the set of strings such that } w \text{ has an even number of 0's and exactly two 1's}\}$.

(You need to construct this language using 7 states)

States x 4 States = States

trap Mergs * 2 lead 7 States

*C 5w/w contains equal number of occurrence of the

Substrings 81 and 10] # (73) -3

4818, 101, 801013 -3]#(1)-7 #01)=2 and #(0)=2

M1_M2

States x 4 States = A * 1 B

D (dead)



M

M2 A B [D

PRED 7 States

* A B A L 2 B a z z

- (c) {w ∈ Σ* | w does not have even length and w does not contain the string 111}
- (d) {w ∈ Σ* | if the number of 1's in w is odd, then the number of 0's in w is also odd}

$\overline{B} \cdot A$

$A = B$ (Whenever A happens, B happens)

think of the Situation, when this rule breaks

Complement: A happens and i does not happen. $A \cdot \overline{B}$ S

good case = (Ans]" this bad case

bad Cass

$\overline{B}$ Everything except

By Cross product

$\overline{b} \cdot M = \overline{a} \cdot \#b$

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Question 3

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Dab_Eba

-

-abalb babab > balbababa

Add $\mapsto$ bbb9 , baba , babb , bbbb $\mapsto$ 1999 , a
aab , abaa , a bab

S & bab , baba a = -

[so 3 = Is

the following languages over Σ .

- $L_1 = \{w : \text{length of } w \text{ is two more than multiple of four}\}$
- $L_2 = \{w : \text{every even position letter in } w \text{ is different from the first letter of } w\}$
- $L_3 = \{w : \text{every } 2k + 1 \text{ position in } w \text{ is b, where } k \geq 0\}$
- $L_4 = \{w : \text{every } 2k + 1 \text{ position in } w \text{ is a, where } k \geq 0\}$

problems.

am for a DFA that recognizes L_1 . (3 points)

am for a DFA that recognizes L_2 . (3 points)

am for a DFA that recognizes L_3 . (3 points)

the "cross product" construction to obtain a DFA for the language $L_2 \cap (L_3 \cup L_4)$, how
it have? (1 point)

strings in $L_2 \cap (L_3 \cup L_4)$. (1 point)

am for a DFA that recognizes $L_2 \cap (L_3 \cup L_4)$ using only four states. (2 points)

ing in $\overline{L_3} \circ L_4$. [Recall: $\overline{L}$ denotes the complement of the language L i.e., $\overline{L} = \Sigma^* - L$] (1

ve justification for your answer. (1 point)

AbS

For each of the languages below, write the regular expression for the given language. You may assume that each language is over the alphabet $\Sigma = \{0, 1\}$.

- (a) $\{w \in \Sigma^* \mid w \text{ contains equal number of 01 and 10}\}$ $82^*3^+12^*\underline{1}+8^+\underline{1}+4$

- (b) $\{w \in \Sigma^* \mid w \text{ do not contain substring ba}\}$

=Egiby

(c) $\{w \in \Sigma^* \mid w \text{ is the set of all strings whose 1st, 3rd, 5th, } \dots, \text{ characters are the same}\}$

(d) $\{w \in \Sigma^* \mid w \text{ starts with 0 and has odd length, or starts with 1 and has even length}\}$

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Question 5

two 1}
th 0}
n w is 1}
at least two 0}

[2

$-2 \downarrow [_{3,3-y}^*, \underline{1}_a$

[=&There exist a $\underline{1}$ in w
followed
which is at most
one BY

notes the complement of the language L i.e.,

Question 6

Convert the following DFA into an equivalent regular expression using the state elimination method. First eliminate q_2 , then q_1 , and finally q_0 . You must show work.

